

MongoDB Schemas

User
profile, hash

Crop
plot, variety

Scan
image, result

Task
field work

Note
observation

Advisory
guidance

OTP
recovery

RefreshToken
session

Repository Layer — `BaseRepository<T>` provides typed CRUD helpers; concrete repos (`UserRepo`, `CropRepo`, `ScanRepo`, `TaskRepo`, `NoteRepo`, `AdvisoryRepo`) add domain-specific queries

DI Container

register repos → services → controllers
single construction point at bootstrap

Service Layer

AuthService
tokens, OTP

UserService
profile ops

CropService
plot CRUD

ScanService
diagnosis

Data Paths

Write Path

validate DTO → service rule
→ repo mutation → model save

Read Path

route params → repo query
→ ownership check → serialize

MongoDB Atlas

MONGO_URI — connection lifecycle, indexes, production datastore

Architecture — Controllers stay thin; services own business rules; repositories isolate persistence mechanics behind `BaseRepository<T>`. The DI container wires everything at bootstrap so each layer depends only on abstractions, not concrete implementations. JWT user context from `req.user` propagates through the service layer for ownership checks.